

AI-SOC-Assistant Evolution & Wins

A concise engineering brief showing how the project matured into a reproducible, explainable, and safety-conscious SOC triage pipeline with a live analyst workflow.

Capability	Professional signal
5-class family classification	Separates Normal, DoS, Probe, R2L, and U2R instead of returning only a binary anomaly flag.
Stable dual-detector triage	Combines supervised attack probability with a training-calibrated Isolation Forest risk signal.
Validated SHAP evidence	Reports real feature values and whether each contribution supports or opposes the predicted class.
Governed ticket generation	Defaults to a deterministic offline renderer and safely validates optional LLM output.
Closed analyst feedback loop	Append-only SQLite reviews promote confirmed false positives into weighted Random Forest retraining.
External generalisation evidence	A zero-tuning UNSW-NB15 transfer benchmark reports the cross-dataset gap rather than hiding it.

Engineering Evolution

1. Robust Feature Encoding and Class Balance

Categorical protocol, service, and flag fields use OneHotEncoder(handle_unknown="ignore"). Class imbalance is handled only on the training fold through deterministic duplication of exact rows, so categorical, binary, and integer telemetry is never interpolated.

Win: Novel categories no longer crash inference, and balanced training data remains valid network telemetry.

2. Stable Anomaly Calibration

Isolation Forest calibration is learned once from training-normal traffic and reused for batch and single-record scoring. The fused score uses the probability of any non-normal class rather than the classifier's maximum confidence.

Win: Alert selection and ticket generation now agree, independent of unrelated test-batch rows.

3. LangGraph Orchestration with Pinned Compatibility

The ReAct agent uses the state_modifier prompt API supported by the pinned LangGraph 0.2.53 stack, with direct provider integrations for Ollama, OpenAI, and Anthropic.

Win: Every supported provider can construct the agent graph without a version-signature failure.

4. Local-First Network Boundary

Template mode makes no network calls, and the default Compose service runs with networking disabled. Ollama keeps prompts on the configured local server; cloud providers are explicitly external. AbuseIPDB and NVD tools remain off unless SOC_ENABLE_THREAT_INTEL=true is deliberately set.

Win: Privacy claims now match the actual deployment boundary and require explicit opt-in for external lookups.

5. Live Ingestion and a Closed Feedback Loop

Delayed NSL-KDD replay and a Kafka-compatible consumer and publisher feed the same scoring runtime as the CLI. Generated tickets persist to SQLite, analyst reviews are append-only, and confirmed false positives re-enter training with an explicit per-row sample weight behind a versioned, atomically written artifact.

Win: The project demonstrates an operating loop rather than a batch script, and model promotion stays deliberate and auditable.

Safety, Explainability, and Evaluation

6. Threat-Intelligence Correctness and Input Safety

IP addresses and service names are validated before use. API calls have timeouts and malformed-response handling. NVD searches use a bounded 119-day publication window and sort results newest first.

Win: Unsafe inputs are rejected, and API failures cannot prevent deterministic incident-ticket generation.

7. Class-Aware SHAP Semantics

Predicted labels are mapped through the fitted model's classes_ ordering, feature schemas must match exactly, and contribution directions are described as supporting, opposing, or neutral for the predicted class.

Win: Explanations remain correct for non-contiguous labels and avoid unsupported causal attack claims.

8. Evidence-Bound GenAI Guardrails

Only allow-listed detector evidence reaches the LLM. Returned tickets must preserve validated detector semantics and evidence, use safe source-bound SPL queries, and omit raw SHAP values. Any malformed, unsafe, or provider-failed response falls back to the deterministic renderer.

Win: The optional LLM acts as a governed presentation layer rather than an authority over incident facts.

9. Protocol-Scoped, Reproducible Evaluation

Hold-out, KDDTest+ cross-distribution, and external UNSW-NB15 transfer evaluations write separate artifacts. Current results are 99.88% hold-out accuracy, 74.40% cross-distribution accuracy, and 58.89% zero-tuning transfer accuracy at 16.02% macro F1. Dataset checksums, supported Python versions, and a shared-code notebook make reruns auditable.

Win: Strong in-distribution performance is presented alongside honest rare-class, distribution-shift, and cross-dataset limits.

10. Explanation-First Analyst Interface

The Streamlit console is built on a single design-system module that defines the colour tokens, type ramp, and render helpers. SHAP drivers appear as signed contribution bars labelled as supporting or opposing the predicted class, while raw SHAP floats stay out of the ticket. Every rendered value is HTML-escaped because source IPs, event ids, and feature names reach the DOM.

Win: The explanation chain is legible to an analyst in one screen, and the interface inherits the same evidence-handling discipline as the pipeline.

Validation snapshot

Clean Python 3.10 to 3.12 dependency installs, 118 automated tests, formatting, lint, dependency consistency, default and no-balancing pipeline runs, and all three evaluation protocols pass. The analyst console is additionally verified by rendering it in a headless browser and asserting that no view raises, that Material icons resolve to glyphs, and that no element overflows, clips, or overlaps. Container build and non-root checks are enforced in CI across Python 3.10-3.12.

Portfolio Positioning

The repository now demonstrates not only model training, but also production-minded evidence handling, network boundaries, dependency hygiene, regression coverage, a closed analyst feedback loop, and honest evaluation across three protocols. Those qualities make the project easier to operate, review, and extend in its next implementation phase.

Updated 12 August 2026